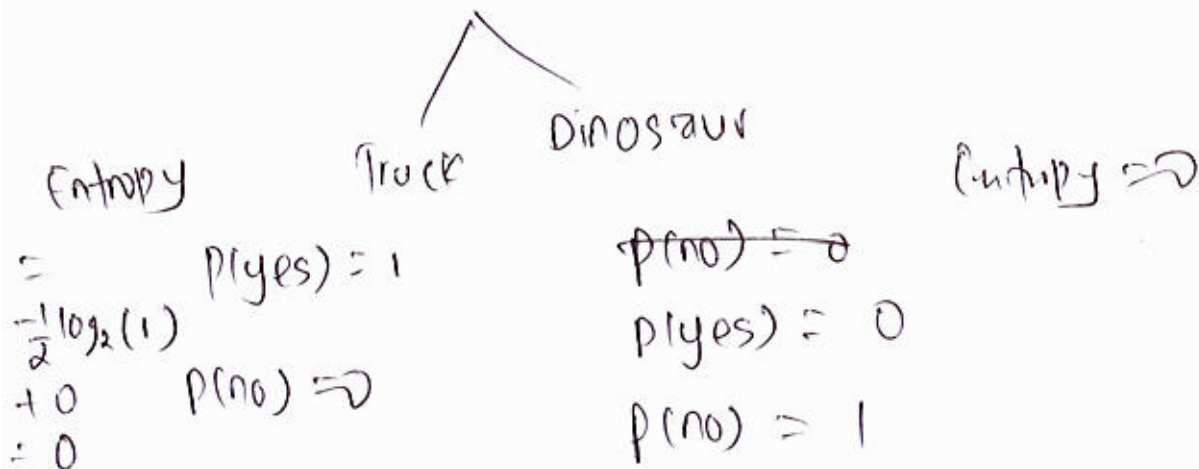


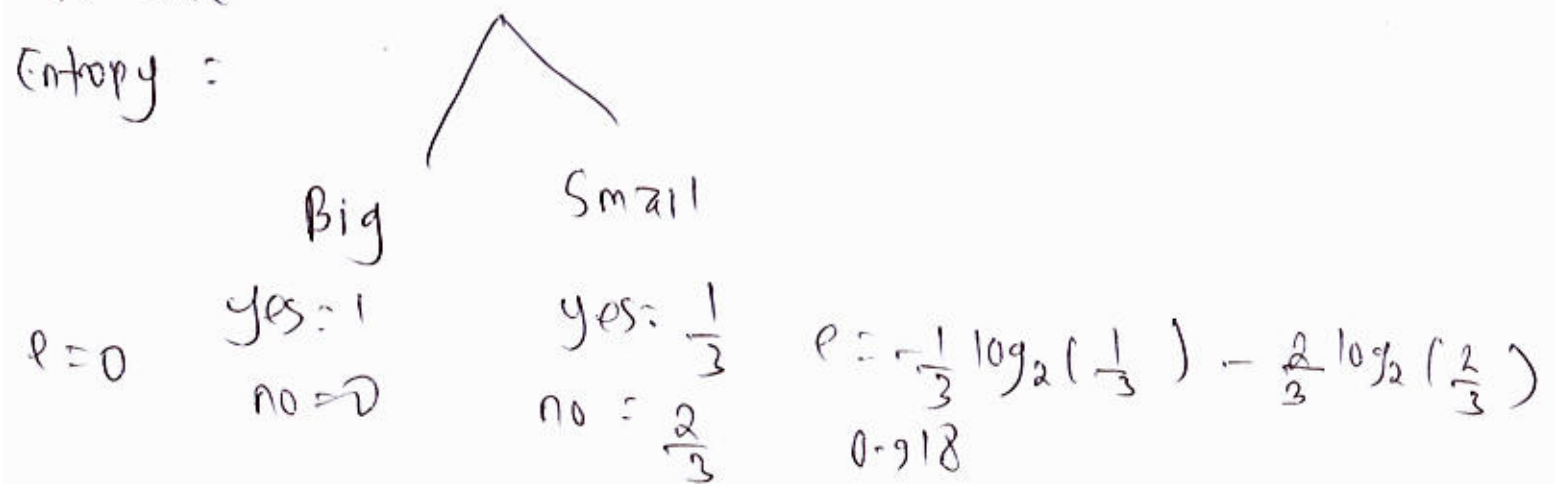
Type . . .



Information gain = 0.971

③ Hence, the maximum information gain comes from splitting the type of toys.

for size .



Information gain = $0.971 - 0 - \frac{3}{5} \cdot 0.918 = 0.020 \text{ bits}$

$$\textcircled{1} \quad p(\overset{\text{yes}}{\text{Is-tur}}) = \frac{3}{5}$$

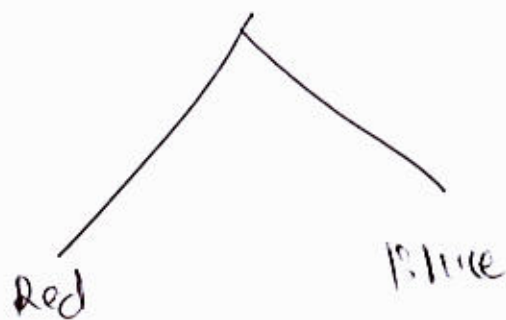
$$p(\text{no}) = \frac{2}{5}$$

$$\begin{aligned} \text{Entropy} &= -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \\ &= 0.971 \text{ bits} \end{aligned}$$

$$\textcircled{2} \quad \therefore p(\text{Red}) = \frac{3}{5} \quad \text{Blue} = \frac{2}{5}$$

~~Information gain~~

~~Size~~ ~~$p(\text{big})$~~ $p(\text{yes}) =$



$$\begin{aligned} \text{Entropy} &= 0.918 \\ \text{yes} &= \frac{2}{3} \\ \text{no} &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{Entropy} &= 1 \\ \text{yes} &= \frac{1}{2} \\ \text{no} &= \frac{1}{2} \end{aligned}$$

$$\text{Information gain} = 0.971 - \frac{3}{5} 0.918 - \frac{2}{5} 1 = 0.020 \text{ bits}$$